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United States Patent	12387427
Kind Code	B2
Date of Patent	August 12, 2025
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Light field camera system and method of setting baseline and convergence distance

Abstract

A light field camera system and method of setting a baseline and a convergence distance of a multiple camera rig employ sample point depths to determine and set values with respect to a three-dimensional (3D) scene. A baseline and a convergence distance of a multiple camera rig is set by determining a plurality of sample point depths representing distances between the multiple camera rig and a plurality of sample points in the 3D scene within a field of view of the multiple camera rig. The convergence distance is set to be an average sample point depth of the plurality of sample point depths. The baseline is then set as a negative inverse of a difference between a minimum sample point depth of the plurality of sample point depths and the convergence distance. The light field camera system includes a multiple camera rig comprising a plurality of cameras.

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Appl. No.:	18/229605
Filed:	August 02, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230410419 A1	Dec. 21, 2023

Related U.S. Application Data

continuation parent-doc WO PCT/US2022/014613 20220131 PENDING child-doc US 18229605
us-provisional-application US 63148587 20210211

Publication Classification

Int. Cl.: G06T17/00 (20060101); H04N13/351 (20180101); H04N23/90 (20230101); H04N23/957 (20230101)

U.S. Cl.:

CPC G06T17/00 (20130101); H04N13/351 (20180501); H04N23/90 (20230101); H04N23/957 (20230101);

Field of Classification Search

CPC: G06T (17/00); H04N (13/351); H04N (23/90); H04N (23/957); H04N (13/359); H04N (13/239); H04N (2013/0081); H04N (13/282); H04N (13/296); H04N (13/243); H04N (13/279); H04N (13/271)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation patent application of and claims priority to International Patent Application No. PCT/US2022/014613, filed Jan. 31, 2022, which claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 63/148,587, filed Feb. 11, 2021, the entirety of each of which is incorporated by reference herein.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) N/A

BACKGROUND

(2) A scene in three-dimensional (3D) space may be viewed from multiple perspectives depending on the viewing angle. In addition, when viewed with stereoscopic vision, multiple views representing different perspectives of the scene may be perceived contemporaneously, effectively creating a sense of depth that may be perceived by a viewer. Multiview displays present images having multiple views to represent how scenes are perceived in the 3D world. A multiview display renders different views contemporaneously to provide a realistic experience to the user. Multiview images may be generated and processed dynamically by software. Capturing multiview images may involve multiple cameras or camera positions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various features of examples and embodiments in accordance with the principles described herein may be more readily understood with reference to the following detailed description taken in conjunction with the accompanying drawings, where like reference numerals designate like structural elements, and in which:

(2) FIG. 1A illustrates a perspective view of a multiview display in an example, according to an embodiment consistent with the principles described herein.

(3) FIG. 1B illustrates displaying a multiview image using a multiview display in an example, according to an embodiment consistent with the principles described herein.

(4) FIG. 2 illustrates capturing a multiview image of a three dimensional (3D) scene in an example, according to an embodiment consistent with the principles described herein.

(5) FIG. 3 illustrates a flow chart of a method of setting a baseline and a convergence distance of a multiple camera rig in an example, according to an embodiment of the principles described herein.

(6) FIG. 4 illustrates ray casting in an example, according to an embodiment of the principles described herein.

(7) FIG. 5 illustrates a block diagram of a light field camera system in an example, according to an embodiment of the principles described herein.

(8) FIG. 6A illustrates a cross-sectional view of a multiview display in an example, according to an embodiment consistent with the principles described herein.

(9) FIG. 6B illustrates a cross-sectional view of a multiview display in another example, according to an embodiment consistent with the principles described herein.

(10) FIG. 6C illustrates a perspective view of a multiview display in an example, according to an embodiment consistent with the principles described herein.

(11) FIG. 7 illustrates a block diagram of a client device in an example, according to an embodiment consistent with the principles described herein.

(12) Certain examples and embodiments may have other features that are one of in addition to and in lieu of the features illustrated in the above-referenced figures. These and other features are detailed below with reference to the above-referenced figures.

DETAILED DESCRIPTION

(13) Examples and embodiments in accordance with the principles described herein provide techniques to set both baseline and convergence distance of a camera rig used to capture multiview images of a three dimensional (3D) scene. In particular, according to various embodiments, the baseline and convergence distance of a multiple camera rig may be determined based on three-dimensional (3D) content captured by a multiple camera rig. In turn, the baseline and a convergence distance may be dynamically adjusted based on the 3D content visible to one or more cameras of the multiple camera rig as the 3D content changes. According to various embodiments, setting the baseline and convergence distance may employ either physical cameras for capturing light field images and video or virtual cameras such as those found in any of a variety of rendering engines (e.g., 3D modeling/animation software, game engines, video editing tools).

(14) As explained below, rather than using a preset baseline and convergence distance or manually adjusted the baseline and the convergence distance, embodiments are directed to modifying these parameters depending on the various depths of a scene within a view of a camera or more particularly one or more cameras of a multiple camera rig. The camera or cameras, in turn, may be either a virtual camera(s) or an actual or physical camera(s). When implemented in a renderer, the virtual camera may be positioned to capture a 3D scene of a portion of a 3D model. A renderer may be a gaming engine, 3D model player, video player, or other software environment that positions a virtual camera to capture a 3D model.

(15) According to various embodiments, a camera generally has a particular location (e.g., coordinates) and a direction to capture a view representing the 3D scene of the 3D model. In this respect, there are a plurality of depths (sample point depths) between the camera and the various surfaces of the 3D scene within the view of the camera. To generate a multiview image, a plurality of cameras or 'multiple camera rig' captures varying, overlapping views of the 3D scene. In some embodiments that employ virtual cameras, the virtual camera (e.g., reference camera) may be replicated (e.g., spawned, copied), to generate the multiple camera rig that captures varying, overlapping views of the 3D scene from, for example, a 3D model. In other embodiments, the multiple camera rig comprises physical or actual cameras configured to capture an image representing different views of a physical or actual 3D scene. In these embodiments, the depths represent actual or physical depths within the physical 3D scene.

(16) FIG. 1A illustrates a perspective view of a multiview display **100** (or multiview mode of a multimode display) in an example, according to an embodiment consistent with the principles described herein. As illustrated in FIG. 1A, the multiview display **100** comprises a screen configured to display a multiview image **110** to be viewed. The multiview display **100** provides different views **112** of the multiview image in different view directions **120** relative to the screen of the multiview display **100**. The view directions **120** are illustrated as arrows extending from the screen in various different principal angular directions; the different views **112** are illustrated as shaded polygonal boxes at the termination of the arrows (i.e., depicting the view directions **120**); and only four views **112** and four view directions **120** are illustrated, all by way of example and not limitation. Note that while the different views **112** are illustrated in FIG. 1A as being above the screen, the views **112** actually appear on or in a vicinity of the screen when the multiview image **110** is displayed on the multiview display **100**. Depiction of the views **112** above the screen of the multiview display **100** in FIG. 1A is only for simplicity of illustration and is meant to represent viewing the multiview display **100** from a respective one of the view directions **120** corresponding to a particular view **112**. As illustrated the multiview display **100** configured to display a multiview

image **110** may be or serve as a display (e.g., a display screen) of a telephone (e.g., mobile telephone, smart phone, etc.), a tablet computer, a laptop computer, a computer monitor of a desktop computer, a camera display, or an electronic display of substantially any other device, according to various embodiments.

(17) A view direction or equivalently a light beam having a direction corresponding to a view direction of a multiview display generally has a principal angular direction given by angular components $\{\theta, \phi\}$, by definition herein. The angular component θ is referred to herein as the ‘elevation component’ or ‘elevation angle’ of the light beam. The angular component ϕ is referred to as the ‘azimuth component’ or ‘azimuth angle’ of the light beam. By definition, the elevation angle θ is an angle in a vertical plane (e.g., perpendicular to a plane of the multiview display screen while the azimuth angle ϕ is an angle in a horizontal plane (e.g., parallel to the multiview display screen plane).

(18) FIG. 1B illustrates displaying a multiview image **110** using a multiview display **100** in an example, according to an embodiment consistent with the principles described herein. The multiview image **110** has a plurality of views **112**. Each of the views **112** corresponds to a different view direction **120** or perspective of scene. The views **112** are rendered for display by a multiview display **100**. As such, each view **112** represents a different viewing angle of the multiview image **110**. The different views **112** therefore have some level of disparity with respect to one another. In some embodiments, a viewer may perceive one view **112** with a right eye while perceiving a different view **112** with a left eye. This allows the viewer to perceive different views simultaneously and may result in stereoscopic vision. In other words, the different views **112** create a three-dimensional (3D) effect.

(19) In some embodiments, as a viewer physically changes a viewing angle with respect to the multiview display **100**, the eyes of the viewer may encounter different views **112** of the multiview image **110** at different points in time as the viewing angle changes. As a result, the viewer may interact with the multiview display **100** by changing the viewing angle to see different views **112** of the multiview image **110**. For example, as the viewer moves to the left, the viewer may see more of the left side of an object in the multiview image **110**. According to various embodiments, the multiview image **110** may have multiple views **112** along a horizontal plane or axis providing a so-called ‘horizontal-parallax-only’ (HPO) 3D multiview image, while in other embodiments, the multiview image **110** may have multiple views **112** along both the horizontal axis and a vertical plane or axis resulting in a so-called ‘full parallax’ 3D multiview image. Thus, as a viewer changes the viewing angle to see different views **112**, the viewer may gain additional visual details within the multiview image **110**. When processed for display, the multiview image **110** is stored as data in a format that records the different views **112**, according to various embodiments.

(20) Herein a ‘two-dimensional display’ or ‘2D display’ is defined as a display configured to provide a view of an image that is substantially the same regardless of a direction from which the image is viewed (i.e., within a predefined viewing angle or range of the 2D display). A conventional liquid crystal display (LCD) found in many smart phones and computer monitors are examples of 2D displays. In contrast herein, a ‘multiview display’ is defined as an electronic display or display system configured to provide different views of a multiview image in or from different view directions contemporaneously from the user's perspective. In particular, the different views **112** may represent different perspective views of a multiview image **110**.

(21) As is described in more detail below, the multiview display **100** may be implemented using a variety of technologies that accommodate the presentation of different image views so that they are perceived contemporaneously. One example of a multiview display is one that employs diffraction gratings to control the principle angular directions of the different views **112**. According to some embodiments, the multiview display **100** may be a light field display, which is one that presents a plurality of light beams of different colors and different directions corresponding to different views. In some examples, the light field display is a so-called ‘glasses free’ three-dimensional (3D)

display that may use diffractive gratings or multibeam elements to provide autostereoscopic representations of multiview images without the need for special eye wear to perceive depth. In some embodiments, the multiview display **100** may require glasses or other eyewear to control which views **112** are perceived by each eye of the user.

(22) In some embodiments, the multiview display **100** is part of a multiview display system that renders multiview images and 2D images. In this respect, the multiview display system may include a plurality of backlights to operate in different modes. For example, the multiview display system may be configured to provide broad-angle emitted light during a 2D mode using a broad-angle backlight. In addition, the multiview display system may be configured to provide directional emitted light during a multiview mode using a multiview backlight having an array of multibeam elements, the directional emitted light comprising a plurality of directional light beams provided by each multibeam element of the multibeam element array. The multiview display system may be configured to time multiplex the 2D and multiview modes using a mode controller to sequentially activate the broad-angle backlight during a first sequential time interval corresponding to the 2D mode and the multiview backlight during a second sequential time interval corresponding to the multiview mode. Directions of directional light beams of the directional light beam may correspond to different view directions of a multiview image.

(23) For example, in 2D mode, the broad-angle backlight may generate images so that the multiview display system operates like a 2D display. By definition, 'broad-angle' emitted light is defined as light having a cone angle that is greater than a cone angle of the view of a multiview image or multiview display. In particular, in some embodiments, the broad-angle emitted light may have a cone angle that is greater than about twenty degrees (e.g., $>\pm 20^\circ$). In other embodiments, the broad-angle emitted light cone angle may be greater than about thirty degrees (e.g., $>\pm 30^\circ$), or greater than about forty degrees (e.g., $>\pm 40^\circ$), or greater than about fifty degrees (e.g., $>\pm 50^\circ$). For example, the cone angle of the broad-angle emitted light may be greater than about sixty degrees (e.g., $>\pm 60^\circ$).

(24) The multiview mode may use a multiview backlight instead of a broad-angle backlight. The multiview backlight may have an array of multibeam elements that scatter light as plurality of directional light beams having principal angular directions that differ from one another. For example, if the multiview display **100** operates in a multiview mode to display a multiview image having four views, the multiview backlight may scatter light into four directional light beams, each directional light beam corresponding to a different view. A mode controller may sequentially switch between 2D mode and multiview mode so that a multiview image is displayed in a first sequential time interval using the multiview backlight and a 2D image is displayed in a second sequential time interval using the broad-angle backlight.

(25) In some embodiments, the multiview display system is configured to guide light in a light guide as guided light. Herein, a 'light guide' is defined as a structure that guides light within the structure using total internal reflection or 'TIR'. In particular, the light guide may include a core that is substantially transparent at an operational wavelength of the light guide. In various examples, the term 'light guide' generally refers to a dielectric optical waveguide that employs total internal reflection to guide light at an interface between a dielectric material of the light guide and a material or medium that surrounds that light guide. By definition, a condition for total internal reflection is that a refractive index of the light guide is greater than a refractive index of a surrounding medium adjacent to a surface of the light guide material. In some embodiments, the light guide may include a coating in addition to or instead of the aforementioned refractive index difference to further facilitate the total internal reflection. The coating may be a reflective coating, for example. The light guide may be any of several light guides including, but not limited to, one or both of a plate or slab guide and a strip guide. The light guide may be shaped like a plate or slab. The light guide may be edge lit by a light source (e.g., light emitting device).

(26) In some embodiments, the multiview display system is configured to scatter out a portion of

the guided light as the directional emitted light using multibeam elements of the multibeam element array, each multibeam element of the multibeam element array comprising one or more of a diffraction grating, a micro-refractive element, and a micro-reflective element. In some embodiments, a diffraction grating of a multibeam element may comprise a plurality of individual sub-gratings. In some embodiments, a micro-reflective element is configured to reflectively couple or scatter out the guided light portion as the plurality of directional light beams. The micro-reflective element may have a reflective coating to control the way guided light is scattered. In some embodiments, the multibeam element comprises a micro-refractive element that is configured to couple or scatter out the guided light portion as the plurality of directional light beams by or using refraction (i.e., refractively scatter out the guided light portion).

(27) FIG. 2 illustrates capturing a multiview image of a three dimensional (3D) scene **200** in an example, according to an embodiment consistent with the principles described herein. As illustrated in FIG. 2, the 3D scene **200** contains various objects (e.g., physical or virtual) such as a tree **202** and a rock **204** on the ground **208**. The tree **202**, the rock **204**, and the ground **208** may be referred to as objects, which together, form at least part of a 3D scene **200**. The multiview image of the 3D scene **200** may be displayed and viewed in a manner like the manner discussed with respect to FIGS. 1A-1B. To capture the multiview image, a camera(s) **210** may be used. In some embodiments, the camera **210** may include one or more physical cameras. For example, a physical camera includes a lens for capturing light and recording the light as an image. Multiple physical cameras may be used to capture different views of the scene to create the multiview image. For example, each physical camera may be spaced apart at a defined distance to allow different perspectives of the objects in the scene to be captured. The distance between the different physical cameras allows for the ability to capture depth of the 3D scene **200**, the same way that the distance between a viewer's eyes allow for 3D vision.

(28) The camera **210** may also represent one or more a virtual (e.g., simulated or hypothetical) cameras as opposed to a physical camera. The 3D scene **200** may be generated using computer graphics techniques that manipulate computer-generated information. In this example, the camera **210** is implemented as a virtual camera having a point of view to capture the 3D scene **200**. The virtual camera may be defined in terms of viewing angle and coordinates within a 3D model. The 3D model may define various objects (e.g., the tree **202**, the rock **204**, and the ground **208**) that are captured by the virtual camera(s).

(29) When generating or capturing views of a scene using the camera **210**, the camera may be configured according to a plane of convergence **230**. A 'plane of convergence' or 'convergence plane' is defined as a plurality of locations where the different views align such that there is little to no disparity between the different views. The plane of convergence **230** occurs in front of the camera **210**. Objects between the camera **210** and the plane of convergence **230** appear closer to the viewer while objects behind the plane of convergence **230** appear further away from the viewer. In this respect, the degree of disparity between the different views increases the further the object is positioned away from the plane of convergence **230**. Objects along the plane of convergence **230** appear in focus with respect to the viewer. The distance between the camera **210** and the plane of convergence **230** is referred to as the convergence distance or convergence offset. As the camera **210** changes position or direction or as the scene changes, the convergence distance is dynamically updated as described herein,

(30) The camera **210** captures a scene that falls within the frustum **220** of the camera **210**. The frustum **220** is shown to have an upper bound and a lower bound that define the viewing angle range of the 3D scene **200**. In FIG. 2, the plane of convergence **230** intersects the bottom of the tree **202** and the back of the tree **202** (relative to the camera **210**). As a result, the bottom of the tree **202** appears to be in focus and will appear as the featured point of interest to a viewer as it appears to be located on the display. In addition, the rock **204** may appear in a foreground in front of the tree **202**.

(31) As used herein, 'disparity' is defined as the difference between at least two views of a

multiview image at corresponding locations. For example, in the context of stereoscopic vision, the left eye and right eye may see the same object but at slightly different locations due to the difference in viewing angles between the eyes. This difference may be quantified as disparity. The change in disparity across the multiview image conveys a sense of depth.

(32) The terms ‘baseline’ or ‘camera baseline’ are defined as the distance between two cameras that capture corresponding views of a multiview image. For example, in the context of stereoscopic vision, the baseline is the distance between the left eye and right eye. A larger baseline leads to increased disparity and may enhance the 3D effect of the multiview image. Scaling the baseline or baseline scaling refers to changing or adjusting the baseline according to a multiplier to either reduce or increase the baseline. Herein, pairs of cameras of a multiple camera rig are spaced apart from one another by a baseline, by definition. In some embodiments, a common baseline is used between each pair of cameras of the multiple camera rig.

(33) Herein ‘convergence distance’ or ‘convergence offset’ refers to the distance between the camera and point along the plane of convergence, by definition. Modifying the convergence offset will change the location of the plane of convergence so as to refocus the multiview image on new objects at a different depth.

(34) Further herein, a ‘3D scene’ refers to a scene containing one or more 3D objects that may exist in physical space or may be virtually represented as a 3D model or 3D environment. A physical 3D scene may be captured by physical cameras while a virtual 3D scene may be captured by virtual cameras.

(35) Further, as used herein, the article ‘a’ is intended to have its ordinary meaning in the patent arts, namely ‘one or more’. For example, ‘a camera’ means one or more cameras and as such, ‘the camera’ means ‘the camera(s)’ herein. Also, any reference herein to ‘top’, ‘bottom’, ‘upper’, ‘lower’, ‘up’, ‘down’, ‘front’, ‘back’, ‘first’, ‘second’, ‘left’ or ‘right’ is not intended to be a limitation herein. Herein, the term ‘about’ when applied to a value generally means within the tolerance range of the equipment used to produce the value, or may mean plus or minus 10%, or plus or minus 5%, or plus or minus 1%, unless otherwise expressly specified. Further, the term ‘substantially’ as used herein means a majority, or almost all, or all, or an amount within a range of about 51% to about 100%. Moreover, examples herein are intended to be illustrative only and are presented for discussion purposes and not by way of limitation.

(36) According to some embodiments of the principles described herein, a method of setting a baseline and a convergence distance of a multiple camera rig is provided. FIG. 3 illustrates a flow chart of a method **300** of setting a baseline and a convergence distance of a multiple camera rig in an example, according to an embodiment of the principles described herein. In some embodiments, the method **300** of setting a baseline and a convergence distance of a multiple camera rig may be used to dynamically adjust both the baseline and convergence distance of the multiple camera rig. For example, baseline adjustment or baseline scaling along with convergence distance adjustment may be implemented in real-time, according to some embodiments.

(37) As illustrated, the method **300** of setting a baseline and a convergence distance of a multiple camera rig comprises determining **310** a set or a plurality of sample point depths. According to various embodiments, the sample point depths represent a collection or plurality of distances between the multiple camera rig and a plurality of sample points in a three dimensional (3D) scene. For example, the 3D scene may be a scene that is within a field of view of the multiple camera rig (e.g., visible to cameras of the multiple camera rig), and the distances may represent distances to various objects or points of interest within the 3D scene.

(38) In some embodiments, determining **310** a plurality of sample point depths may comprise performing ray casting within the 3D scene. For example, a grid of ray casts may be generated outwards from the multiple camera rig toward the 3D scene. Hit distances to various colliders in the 3D scene are then recorded for the ray casts in the ray cast grid. The hit distances correspond to the depths of the various colliders within the 3D scene.

(39) FIG. 4 illustrates ray casting in an example, according to an embodiment of the principles described herein. As illustrated, a 3D scene **402** includes a plurality of objects **404** while a multiple camera rig **410** is positioned to capture an multiview image the 3D scene **402**. Ray casting comprises generating a plurality or grid of rays **420** and directing the rays into the 3D scene **402** where individual rays **420** encounter points on the objects **404**, as illustrated by arrow in FIG. 4. As illustrated, hit distances are lengths of the rays **420** extending between an origin point at the multiple camera rig **410** and a point at which the ray **420** encounters (i.e., terminates on) a particular object **404**. The hit distances for the grid of rays provided by ray casting then determines **310** the plurality of sample point depths (or distances). FIG. 4 also illustrates a baseline **b** between adjacent cameras of the multiple camera rig **410**.

(40) In other embodiments, determining **310** a plurality of sample point depths may comprise computing the depths from a disparity map of the 3D scene. The disparity map may be provided along with an image of the scene, in some embodiments. In other embodiments, the disparity map may be computed from a visual disparity between images recorded by different cameras of the multiple camera rig. In particular, computing the depths may comprise using image disparity between images captured by the different cameras of the multiple camera rig. For example, a depth buffer associated with images of the 3D scene captured by cameras of the multiple camera rig may be interrogated. Computing the depths may further comprise forming a disparity map of the 3D scene from the image disparity, in some embodiments.

(41) In yet other embodiments, determining **310** a plurality of sample point depths may comprise using a depth sensor to measure distances between the multiple camera rig and objects within the 3D scene. In various embodiments, the objects may correspond to sample points of the plurality of sample points and the depth sensor may comprise any of a variety of depth sensors. For example, the depth sensor may be a laser distance sensor including, but not limited to, a laser detection and ranging (LIDAR) system. In another example, the depth sensor may be a time-of-flight distance sensor. In yet another example, a ranging or depth measuring system that employs either sound waves (e.g., a sound navigation and ranging or ‘SONAR’ system) or structured light. For example, an image with different colors at different heights may be projected onto the scene, and then an image captured of the scene by a camera may be used by an algorithm to generate a depth map by assigning depths based on the colors of each pixel. Even robots with 3D tracking may be used to effectively explore or ‘roll’ over a landscape and record heights or distances, etc., may be used to determine the sample point depths.

(42) As illustrated in FIG. 3, the method **300** of setting a baseline and a convergence distance of a multiple camera rig further comprises setting **320** the convergence distance to be an average depth of the plurality of sample point depths. In particular, an average of sample point depths of a plurality of the sample point depths is computed. The convergence distance is then set to be equal to the computed average of the sample point depths. In some embodiments, the plurality of sample point depths comprises depths of all of the sample points. In other embodiments, the plurality of sample point depths may comprise a selection of the sample point depths that is smaller than a total number of the sample points, e.g., the set may include only sample points that are considered to be relevant or important, e.g., sample points associated with major objects or colliders in the 3D scene. In another embodiment (e.g., for large 3D models), the plurality of sample points depths may be determined for a subset of the plurality of vertices (e.g., every other or every third vertex). Referring again to FIG. 4, an average sample depths is illustrated as an average ray cast hit distance.

(43) In some embodiments, the average depth of the plurality of sample points is a weighted average. According to various embodiments, the weighted average S may be computed using equation (1)

$$(44) \quad S = \frac{1}{n} \cdot \text{Math.}_{i=1}^n w_i \cdot \text{Math. } s_i \quad (1)$$

where n is a number of sample points in the plurality of sample points, s are the individual sample points of the plurality of sample points, and $w_{sub.i}$ are the weights of the weighted average S . (45) According to some embodiments, the weights $w_{sub.i}$ of the weighted average S may be selected or assigned based on a distance of the various sample points $s_{sub.i}$ from a predetermined location or a particular object within the 3D scene. For example, the weights $w_{sub.i}$ may be selected or determined based on a distance of an $i_{sup.th}$ sample point $s_{sub.i}$ from a center of the scene. In another example, the weights $w_{sub.i}$ may be selected or assigned based on a distance of the $i_{sup.th}$ sample point $s_{sub.i}$ from a point of focus (focal point) of the cameras within the multiple camera rig. In yet another example, the weights $w_{sub.i}$ may be selected or assigned based on a distance of the $i_{sup.th}$ sample point $s_{sub.i}$ from a particular sample point s^* (e.g., a sample point on or associated with an object of interest within the 3D scene).

(46) In some embodiments, the weights $w_{sub.i}$ of the weighted average S may decrease as a function of the distance, e.g., a distance from the scene center, focal point or particular sample point s^* . In some embodiments, the decrease of the weights $w_{sub.i}$ may have either linear distribution as function of distance or a non-linear function (e.g., an exponential distribution). In some embodiments, weights $w_{sub.i}$ of the weighted average S may be assigned according to a gaussian distribution. The gaussian distribution may be centered on the scene center, the focal point, or the particular sample point s^* , for example.

(47) Referring again to FIG. 3, the method **300** of setting a baseline and a convergence distance of a multiple camera rig illustrated in FIG. 3 further comprises determining **330** a minimum sample point depth of the plurality of sample point depths. Herein, the minimum sample point depth is generally defined as the sample point having the minimum depth or distance from cameras of the multiple camera rig. The minimum sample point depth may be determined **330** by examining the plurality of sample point depths and finding or identifying the sample point having the minimum value, for example. In other examples, the minimum sample point depth may be determined **330** by identifying a group of sample points having a lowest depth (or distance) and then setting the minimum sample point depth to be equal to an average of the sample point depths in the group of lowest depth sample points. In some examples, the lowest depth sample point group may comprise a percentage of the sample points having a lowest depth or distance, e.g., about five percent (5%) or about ten percent (10%) of the sample points with the lowest depths or distances. Referring back to FIG. 4, a minimum sample point depth $z_{sub.min}$ representing a minimum hit distance of rays **420** is illustrated.

(48) Returning again to FIG. 3, according to various embodiments, the method **300** of setting a baseline and a convergence distance of a multiple camera rig further comprises setting **340** the baseline to be a negative inverse of a difference between the minimum sample point depth and the convergence distance, as given by equation (2),

$$(49) \quad b = -\frac{1}{(z_{m.Min.n} - D_{conv})} \quad (2)$$

where b is the baseline, $z_{sub.min}$ is the minimum sample point depth, and $D_{sub.conv}$ is the convergence distance, provided as described above.

(50) According to some embodiments, the multiple camera rig may comprise a plurality of virtual cameras. For example, the 3D scene may be a 3D model and the plurality of virtual cameras may be cameras associated with or used to image the 3D model. In some embodiments, a virtual camera of the virtual camera plurality may be a virtual camera that is managed by a renderer. For example, the multiple camera rig may be associated with a renderer that uses a virtual camera of the multiple camera rig to capture a virtual 3D scene. In another embodiment, the multiple camera rig may comprise a plurality of physical cameras. For example, the 3D scene may be or represent a physical scene that is imaged by cameras of the plurality of physical cameras. In yet another embodiment, the multiple camera rig may comprise a camera (e.g., one or more cameras) that is moved between a plurality of locations to capture images that form the 3D scene. In some embodiments, the sample

point depths may represent a depth or distance with respect to one camera of the multiple camera rig (e.g., a reference camera), while in other embodiments, the sample point depths may be distances with respect to the multiple camera rig as a whole. As mentioned above, FIG. 4 also illustrates the baseline **b** between adjacent cameras of the multiple camera rig **410**.

(51) In other embodiments according to the principles described herein, a light field camera system is provided. In some embodiments, the light field camera system may have or provide automatic baseline and convergence distance determination. Figure illustrates a block diagram of a light field camera system **500** in an example, according to an embodiment of the principles described herein. As illustrated in FIG. 5, the light field camera system **500** comprises a multiple camera rig **510**. The multiple camera rig **510** comprises a plurality of cameras separated from one another by a baseline **b**, as illustrated.

(52) The light field camera system **500** further comprises a processor **520** and a memory **530**. The memory **530** is configured to store instructions that when executed by the processor **520**, implement determining a set or plurality of sample point depths representing distances between the multiple camera rig and a plurality of sample points in a three dimensional (3D) scene within a field of view of the multiple camera rig. In some embodiments, determining the plurality of sample point depths may be substantially similar to determining **310** the sample point depths, as described above with respect to the method **300** of setting a baseline and a convergence distance of a multiple camera rig.

(53) In particular, in some embodiments, the plurality of sample point depths may be determined **310** one or more of from a depth map of an image representing the 3D scene, using a disparity map to compute the depths, and performing ray casting within the 3D scene to determine the sample point depths within the 3D scene. In other embodiments, the plurality of sample point depths may be determined **310** using a depth sensor to measure distances between the multiple camera rig and objects within the 3D scene, the objects corresponding to sample points of the plurality of sample points. In some embodiments, the average depth of the plurality of sample point depths is a weighted average, weights of the weighted average being assigned according to a decreasing function of distance from a focal point of the 3D scene.

(54) In some embodiments, a camera of the plurality of cameras is a virtual camera, the 3D scene being a 3D model. In some embodiments, cameras of the plurality of cameras of the multiple camera rig comprises physical cameras, the 3D scene representing a physical scene imaged by the physical cameras.

(55) When executed by the processor **520** illustrated in FIG. 5, instructions stored in memory **530** further implement setting a convergence distance and baseline of the multiple camera rig. The convergence distance may be an average depth of the plurality of sample point depths and the baseline may be a negative inverse of a minimum sample point depth minus the convergence distance, according to various embodiments. In some embodiments, setting a convergences distance and baseline may be substantially similar to setting **320** a convergence distance and setting **340** a baseline of the above-described method **300** of setting a baseline and a convergence distance of a multiple camera rig.

(56) In some embodiments, an application executed by the processor **520** may generate a 3D model using computer graphics techniques for 3D modeling. A 3D model is a mathematical representation of various surfaces and textures of different objects and may include the spatial relationship between the objects. The application may include a renderer that generates and updates the 3D model depending on user input. The user input may involve navigating through the 3D model by clicking or dragging a cursor, pressing direction buttons, converting the user's physical location to a virtual location within the 3D model, etc. The 3D model may be loaded into the memory **530** and updated thereafter. The 3D model may be converted into multiview images revealing a window into the 3D model. The window may be defined by a plurality of virtual cameras, the multiple camera rig **510** having coordinates and directions within the 3D model. The baseline and

convergence distance of the virtual cameras may be dynamically updated depending on movements of the virtual cameras or the 3D scene changing, in some embodiments.

(57) In some embodiments (e.g., as illustrated in FIG. 5), the light field camera system **500** further comprises a multiview display **540**. In these embodiments, the convergence distance may correspond to a zero disparity plane of the multiview display **540**.

(58) FIG. 6A illustrates a cross-sectional view of a multiview display **600** in an example, according to an embodiment consistent with the principles described herein. FIG. 6B illustrates a cross-sectional view of a multiview display **600** in another example, according to an embodiment consistent with the principles described herein. In particular, FIG. 6A illustrates the multiview display **600** during or according to a first or two-dimensional (2D) mode. FIG. 6B illustrates the multiview display **600** during or according to a second or multiview mode. FIG. 6C illustrates a perspective view of a multiview display **600** in an example, according to an embodiment consistent with the principles described herein. The multiview display **600** is illustrated in FIG. 6C during the multiview mode, by way of example and not limitation. Further, the 2D and multiview modes may be time-multiplexed in time-sequential or time-interlaced manner to provide the 2D and multiview modes in alternating first and second time intervals (e.g., alternating between FIGS. 6A and 6B), according to various embodiments. As such, the multiview display **600** may also be referred to as a 'time-multiplexed, mode-switching' multiview display.

(59) As illustrated, the multiview display **600** is configured to provide or emit light as emitted light **602**. The emitted light **602** may be used to illuminate and provide images using the multiview display **600**, according to various examples and embodiments. For example, the emitted light **602** may be used to illuminate an array of light valves (e.g., light valves **630**, described below) of the multiview display **600**.

(60) According to various embodiments, directional emitted light **602''** may be provided during the multiview mode comprising a plurality of directional light beams having directions corresponding to different view directions of a multiview image. Conversely, during the 2D mode, broad-angle emitted light **602'** may be provided that is largely non-directional and further generally has a cone angle that is greater than a cone angle of a view of the multiview image or multiview display associated with the multiview display **600**, according to various embodiments. The broad-angle emitted light **602'** is illustrated in FIG. 6A as dashed arrows for ease of illustration. However, the dashed arrows representing the broad-angle emitted light **602'** are not meant to imply any particular directionality of the emitted light **602**, but instead merely represent the emission and transmission of light, e.g., from the multiview display **600**. Similarly, FIGS. 6B and 6C illustrate the directional light beams of the directional emitted light **602''** as a plurality of diverging arrows. The directional light beams may be or represent a light field, in various embodiments.

(61) As illustrated in FIGS. 6A-6C, the time-multiplexed multimode display **600** comprises a broad-angle backlight **610**. The illustrated broad-angle backlight **610** has a planar or substantially planar light-emitting surface configured to provide the broad-angle emitted light **602'** (e.g., see FIG. 6A). According to various embodiments, the broad-angle backlight **610** may be substantially any backlight having a light-emitting surface **610'** configured to provide light to illuminate an array of light valves of a display. For example, the broad-angle backlight **610** may be a direct-emitting or directly illuminated planar backlight. Direct-emitting or directly illuminated planar backlights include, but are not limited to, a backlight panel employing a planar array of cold-cathode fluorescent lamps (CCFLs), neon lamps or light emitting diodes (LEDs) configured to directly illuminate the planar light-emitting surface **610'** and provide the broad-angle emitted light **602'**. An electroluminescent panel (ELP) is another non-limiting example of a direct-emitting planar backlight. In other examples, the broad-angle backlight **610** may comprise a backlight that employs an indirect light source. Such indirectly illuminated backlights may include, but are not limited to, various forms of edge-coupled or so-called 'edge-lit' backlights.

(62) The multiview display **600** illustrated in FIGS. 6A-6C further comprises a multiview backlight

620. As illustrated, the multiview backlight **620** comprises an array of multibeam elements **622**. Multibeam elements **622** of the multibeam element array are spaced apart from one another across the multiview backlight **620**, according to various embodiments. Differing types of multibeam elements **622** may be utilized in the multiview backlight **620** including, but limited to, active emitters and various scattering elements. According to various embodiments, each multibeam element **622** of the multibeam element array is configured to provide a plurality of directional light beams having directions corresponding to different view directions of a multiview image during a multiview mode.

(63) In some embodiments (e.g., as illustrated), the multiview backlight **620** further comprises a light guide **624** configured to guide light as guided light. The light guide **624** may be a plate light guide, in some embodiments. According to various embodiments, the light guide **624** is configured to guide the guided light along a length of the light guide **624** according to total internal reflection. A general propagation direction of the guided light within the light guide **624** is illustrated by a bold arrow in FIG. **6B**. In some embodiments, the guided light may be guided in the propagation direction at a non-zero propagation angle and may comprise collimated light that is collimated according to a predetermined collimation factor σ , as illustrated in FIG. **6B**.

(64) In embodiments that include the light guide **624**, a multibeam element **622** of the multibeam element array may be configured to scatter out a portion of the guided light from within the light guide **624** and to direct the scattered out portion away from an emission surface to provide the directional emitted light **602''**, as illustrated in FIG. **6B**. For example, the guided light portion may be scattered out by the multibeam element **622** through the first surface. Further, as illustrated in FIGS. **6A-6C**, a second surface of the multiview backlight **620** opposite to the first surface may be adjacent to the planar light-emitting surface of the broad-angle backlight **610**, according to various embodiments. Further, the multiview backlight **620** may be substantially transparent (e.g., in at least the 2D mode) to allow the broad-angle emitted light **602'** from the broad-angle backlight **610** to pass or be transmitted through a thickness of the multiview backlight **620**, as illustrated in FIG. **6A** by the dashed arrows that originate at the broad-angle backlight **610** and subsequently pass through the multiview backlight **620**.

(65) In some embodiments (e.g., as illustrated in FIGS. **6A-6C**), the multiview backlight **620** may further comprise a light source **626**. As such, the multiview backlight **620** may be an edge-lit backlight, for example. According to various embodiments, the light source **626** is configured to provide the light to be guided within light guide **624**. In various embodiments, the light source **626** may comprise substantially any source of light (e.g., optical emitter) including, but not limited to, one or more light emitting diodes (LEDs) or a laser (e.g., laser diode). In some embodiments, the light source **626** may comprise an optical emitter configured to produce a substantially monochromatic light having a narrowband spectrum denoted by a particular color. In particular, the color of the monochromatic light may be a primary color of a particular color space or color model (e.g., a red-green-blue (RGB) color model). In other examples, the light source **626** may be a substantially broadband light source configured to provide substantially broadband or polychromatic light. For example, the light source **626** may provide white light. In some embodiments, the light source **626** may comprise a plurality of different optical emitters configured to provide different colors of light. The different optical emitters may be configured to provide light having different, color-specific, non-zero propagation angles of the guided light corresponding to each of the different colors of light. As illustrated in FIG. **6B**, activation of the multiview backlight **620** may comprise activating the light source **626**.

(66) According to some embodiments (e.g., as illustrated in FIGS. **6A-6C**), multibeam elements **622** of the multibeam element array may be located at the first surface of the light guide **624** (e.g., adjacent to the first surface of the multiview backlight **620**). In other embodiments (not illustrated), the multibeam elements **622** may be located within the light guide **624**. In yet other embodiments (not illustrated), the multibeam elements **622** may be located at or on the second surface of the light

guide **624** (e.g., adjacent to the second surface of the multiview backlight **620**). Further, a size of the multibeam element **622** is comparable to a size of a light valve of the multiview display **600**. In some embodiments, the size of the multibeam element **622** may be between about one quarter and two times the light valve size.

(67) As mentioned above and also as illustrated in FIGS. **6A-6C**, the multiview display **600** further comprises an array of light valves **630**. In various embodiments, any of a variety of different types of light valves may be employed as the light valves **630** of the light valve array including, but not limited to, one or more of liquid crystal light valves, electrophoretic light valves, and light valves based on or employing electrowetting. Further, as illustrated, there may be one unique plurality of light valves **630** for each multibeam element **622** of the array of multibeam elements. The unique plurality of light valves **630** may correspond to a multiview pixel of the time-multiplexed multimode display **600**, for example. According to some embodiments, the comparable sizes of the multibeam element **622** and the light valve **630** may be chosen to reduce, or in some examples to minimize, dark zones between views of the multiview display, while at the same time reducing, or in some examples minimizing, an overlap between views of the multiview display or equivalent of the multiview image.

(68) According to various embodiments, the multibeam elements **622** of the multiview backlight **620** may comprise any of a number of different structures configured to scatter out a portion of the guided light. For example, the different structures may include, but are not limited to, diffraction gratings, micro-reflective elements, micro-refractive elements, or various combinations thereof. In some embodiments, the multibeam element **622** comprising a diffraction grating is configured to diffractively couple or scatter out the guided light portion as the directional emitted light **602** comprising a plurality of directional light beams having the different principal angular directions. In some embodiments, a diffraction grating of a multibeam element may comprise a plurality of individual sub-gratings. In other embodiments, the multibeam element **622** comprising a micro-reflective element is configured to reflectively couple or scatter out the guided light portion as the plurality of directional light beams, or the multibeam element **622** comprising a micro-refractive element is configured to couple or scatter out the guided light portion as the plurality of directional light beams by or using refraction (i.e., refractively scatter out the guided light portion).

(69) In some embodiments, the light field camera system **500** of FIG. **5** may be implemented in or using a client device. FIG. **7** illustrates a block diagram of a client device **700** in an example, according to an embodiment consistent with the principles described herein. The light field camera system **500** may comprise the client device **700**, for example. For example, the processor **520** and memory **530** of the light field camera system **500** may be part of the client device **700**.

(70) As illustrated, the client device **700** comprises a system of components that carry out various computing operations for a user of the client device **700**. The client device **700** may be a laptop, tablet, smart phone, touch screen system, intelligent display system, or other client device. The client device **700** may include various components such as, for example, a processor(s) **710**, a memory **720**, input/output (I/O) component(s) **730**, a display **740**, and potentially other components. These components may couple to a bus **750** that serves as a local interface to allow the components of the client device **700** to communicate with each other. While the components of the client device **700** are shown to be contained within the client device **700**, it should be appreciated that at least some of the components may couple to the client device **700** through an external connection. For example, components may externally plug into or otherwise connect with the client device **700** via external ports, sockets, plugs, or connectors.

(71) A processor **710** may be a central processing unit (CPU), graphics processing unit (GPU), any other integrated circuit that performs computing processing operations, or any combination thereof. The processor(s) **710** may include one or more processing cores. The processor(s) **710** comprises circuitry that executes instructions. Instructions include, for example, computer code, programs, logic, or other machine-readable instructions that are received and executed by the processor(s) **710**.

to carry out computing functionality that are embodied in the instructions. The processor(s) **710** may execute instructions to operate on data. For example, the processor(s) **710** may receive input data (e.g., an image), process the input data according to an instruction set, and generate output data (e.g., a processed image). As another example, the processor(s) **710** may receive instructions and generate new instructions for subsequent execution. The processor **710** may comprise the hardware to implement a graphics pipeline that renders output from a renderer. For example, the processor(s) **710** may comprise one or more GPU cores, vector processors, scaler processes, or hardware accelerators.

(72) The memory **720** may include one or more memory components. The memory **720** is defined herein as including either or both of volatile and nonvolatile memory. Volatile memory components are those that do not retain information upon loss of power. Volatile memory may include, for example, random access memory (RAM), static random access memory (SRAM), dynamic random access memory (DRAM), magnetic random access memory (MRAM), or other volatile memory structures. System memory (e.g., main memory, cache, etc.) may be implemented using volatile memory. System memory refers to fast memory that may temporarily store data or instructions for quick read and write access to assist the processor(s) **710**.

(73) Nonvolatile memory components are those that retain information upon a loss of power. Nonvolatile memory includes read-only memory (ROM), hard disk drives, solid-state drives, USB flash drives, memory cards accessed via a memory card reader, floppy disks accessed via an associated floppy disk drive, optical discs accessed via an optical disc drive, magnetic tapes accessed via an appropriate tape drive. The ROM may comprise, for example, a programmable read-only memory (PROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), or other like memory device. Storage memory may be implemented using nonvolatile memory to provide long term retention of data and instructions.

(74) The memory **720** may refer to the combination of volatile and nonvolatile memory used to store instructions as well as data. For example, data and instructions may be stored in nonvolatile memory and loaded into volatile memory for processing by the processor(s) **710**. The execution of instructions may include, for example, a compiled program that is translated into machine code in a format that can be loaded from nonvolatile memory into volatile memory and then run by the processor **710**, source code that is converted in suitable format such as object code that is capable of being loaded into volatile memory for execution by the processor **710**, or source code that is interpreted by another executable program to generate instructions in volatile memory and executed by the processor **710**, etc. Instructions may be stored or loaded in any portion or component of the memory **720** including, for example, RAM, ROM, system memory, storage, or any combination thereof.

(75) While the memory **720** is shown as being separate from other components of the client device **700**, it should be appreciated that the memory **720** may be embedded or otherwise integrated, at least partially, into one or more components. For example, the processor(s) **710** may include onboard memory registers or cache to perform processing operations.

(76) I/O component(s) **730** include, for example, touch screens, speakers, microphones, buttons, switches, dials, camera, sensors, accelerometers, or other components that receive user input or generate output directed to the user. I/O component(s) **730** may receive user input and convert it into data for storage in the memory **720** or for processing by the processor(s) **710**. I/O component(s) **730** may receive data outputted by the memory **720** or processor(s) **710** and convert them into a format that is perceived by the user (e.g., sound, tactile responses, visual information, etc.). The I/O components **730** may include one or more physical cameras coupled to the client device. The client device **700** may control the baseline of the cameras as well as their ability to focus.

(77) A specific type of I/O component **730** is a display **740**. The display **740** may include a

multiview display (e.g., multiview display **100**), a multiview display combined with a 2D display, or any other display that presents images. A capacitive touch screen layer serving as an I/O component **730** may be layered within the display to allow a user to provide input while contemporaneously perceiving visual output. The processor(s) **710** may generate data that is formatted as an image for presentation on the display **740**. The processor(s) **710** may execute instructions to render the image on the display for being perceived by the user.

(78) The bus **750** facilitates communication of instructions and data between the processor(s) **710**, the memory **720**, the I/O component(s) **730**, the display **740**, and any other components of the client device **700**. The bus **750** may include address translators, address decoders, fabric, conductive traces, conductive wires, ports, plugs, sockets, and other connectors to allow for the communication of data and instructions.

(79) The instructions within the memory **720** may be embodied in various forms in a manner that implements at least a portion of the software stack. For example, the instructions may be embodied as an operating system **722**, an application(s) **724**, a device driver (e.g., a display driver **726**), firmware (e.g., display firmware **728**), or other software components. The operating system **722** is a software platform that supports the basic functions of the client device **700**, such as scheduling tasks, controlling I/O components **730**, providing access to hardware resources, managing power, and supporting applications **724**.

(80) An application(s) **724** executes on the operating system **722** and may gain access to hardware resources of the client device **700** via the operating system **722**. In this respect, the execution of the application(s) **724** is controlled, at least in part, by the operating system **722**. The application(s) **724** may be a user-level software program that provides high-level functions, services, and other functionality to the user. In some embodiments, an application **724** may be a dedicated ‘app’ downloadable or otherwise accessible to the user on the client device **700**. The user may launch the application(s) **724** via a user interface provided by the operating system **722**. The application(s) **724** may be developed by developers and defined in various source code formats. The applications **724** may be developed using a number of programming or scripting languages such as, for example, C, C++, C#, Objective C, Java®, Swift, JavaScript, Perl, PHP, Visual Basic®, Python®, Ruby, Go, or other programming languages. The application(s) **724** may be compiled by a compiler into object code or interpreted by an interpreter for execution by the processor(s) **710**. The application **724** may include a renderer or other graphics rendering engine.

(81) Device drivers such as, for example, the display driver **726**, include instructions that allow the operating system **722** to communicate with various I/O components **730**. Each I/O component **730** may have its own device driver. Device drivers may be installed such that they are stored in storage and loaded into system memory. For example, upon installation, a display driver **726** translates a high-level display instruction received from the operating system **722** into lower level instructions implemented by the display **740** to display an image.

(82) Firmware, such as, for example, display firmware **728**, may include machine code or assembly code that allows an I/O component **730** or display **740** to perform low-level operations. Firmware may convert electrical signals of particular component into higher level instructions or data. For example, display firmware **728** may control how a display **740** activates individual pixels at a low level by adjusting voltage or current signals. Firmware may be stored in nonvolatile memory and executed directly from nonvolatile memory. For example, the display firmware **728** may be embodied in a ROM chip coupled to the display **740** such that the ROM chip is separate from other storage and system memory of the client device **700**. The display **740** may include processing circuitry for executing the display firmware **728**.

(83) The operating system **722**, application(s) **724**, drivers (e.g., display driver **726**), firmware (e.g., display firmware), and potentially other instruction sets may each comprise instructions that are executable by the processor(s) **710** or other processing circuitry of the client device **700** to carry out the functionality and operations discussed above. Although the instructions described herein may

be embodied in software or code executed by the processor(s) **710** as discussed above, as an alternative, the instructions may also be embodied in dedicated hardware or a combination of software and dedicated hardware. For example, the functionality and operations carried out by the instructions discussed above may be implemented as a circuit or state machine that employs any one of or a combination of a number of technologies. These technologies may include, but are not limited to, discrete logic circuits having logic gates for implementing various logic functions upon an application of one or more data signals, application specific integrated circuits (ASICs) having appropriate logic gates, field-programmable gate arrays (FPGAs), or other components, etc.

(84) In some embodiments of the principles described a non-transitory, computer-readable storage medium storing executable instructions that, when executed by a processor of a computer system, performs operations of determining a baseline and convergence distance of a multiple camera rig is provided. In particular, instructions that carry out the functionality and operations discussed above may be embodied in a non-transitory, computer-readable storage medium. For example, some embodiments may be directed to a non-transitory, computer-readable storage medium storing executable instructions that, when executed by a processor (e.g., processor **710**) of a computing system (e.g., the client device **700**) cause the processor to perform various functions discussed above, including various operations to dynamically and automatically update a convergence distance or baseline of a multiple camera rig.

(85) In particular, operations performed by processor executing instructions stored in the non-transitory, computer-readable storage medium may comprise determining a set or plurality of sample point depths representing distances between the multiple camera rig and a plurality of sample points in a three dimensional (3D) scene within a field of view of the multiple camera rig, the convergence distance being set as an average depth of the plurality of sample point depths. The operations may further comprise determining a minimum sample point depth of the plurality of sample point depths, the baseline being set as a negative inverse of a difference between the minimum sample point depth and the convergence distance. In some embodiments, determining the plurality of sample point depths comprises one or more of from a depth map of an image representing the 3D scene, using a disparity map to compute the depths, performing ray casting within the 3D scene to determine the sample point depths within the 3D scene, and using a depth sensor to measure distances between the multiple camera rig and objects within the 3D scene, the objects corresponding to sample points of the plurality of sample points. In some embodiments, the average depth of the plurality of sample point depths is a weighted average, weights of the weighted average being assigned according to a decreasing function of distance from a focal point of the 3D scene.

(86) Herein, a 'non-transitory, computer-readable storage medium' is defined as any medium that can contain, store, or maintain the instructions described herein for use by or in connection with an instruction execution system. For example, the non-transitory, computer-readable storage medium may store instructions for use by or in connection with the light field camera system **500** or the client device **700**. Further, the non-transitory, computer-readable storage medium may or may not be part of the client device **700** described above (e.g., part of the memory **720**). The instructions stored by the non-transitory, computer-readable storage medium may include, but are not limited to, statements, code, or declarations that may be fetched from the non-transient computer-readable medium and executed by processing circuitry (e.g., the processor **520** or the processor(s) **710**). Further, the term 'non-transitory, computer-readable storage medium' explicitly excludes transitory medium including, for example, carrier waves, by definition herein.

(87) According to various embodiments, the non-transitory, computer-readable medium may comprise any one of many physical media such as, for example, magnetic, optical, or semiconductor media. More specific examples of a suitable non-transitory, computer-readable medium may include, but are not limited to, magnetic tapes, magnetic floppy diskettes, magnetic hard drives, memory cards, solid-state drives, USB flash drives, or optical discs. Also, the non-

transitory, computer-readable medium may be a random access memory (RAM) including, for example, static random access memory (SRAM) and dynamic random access memory (DRAM), or magnetic random access memory (MRAM). In addition, the non-transitory, computer-readable medium may be a read-only memory (ROM), a programmable read-only memory (PROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), or other type of memory device.

(88) The client device **700** may perform any of the operations or implement the functionality described above. For example, the process flows discussed above may be performed by the client device **700** that executes instructions and processes data. While the client device **700** is shown as a single device, embodiments are not so limited. In some embodiments, the client device **700** may offload processing of instructions in a distributed manner such that a plurality of other client devices **700** or other computing devices that operate together to execute instructions that may be stored or loaded in a distributed arranged. For example, at least some instructions or data may be stored, loaded, or executed in a cloud-based system that operates in conjunction with the client device **700**.

(89) Thus, there have been described examples and embodiments of setting a baseline and a convergence distance of a multiple camera rig with application to a light field camera system. In some embodiments, the baseline and convergence distance may be determined dynamically or in real-time based on the depths of points within a camera view. It should be understood that the above-described examples are merely illustrative of some of the many specific examples that represent the principles described herein. Clearly, those skilled in the art can readily devise numerous other arrangements without departing from the discussion presented herein.

Claims

1. A method of setting a baseline and a convergence distance of a multiple camera rig, the method comprising: determining a plurality of sample point depths representing distances between the multiple camera rig and a plurality of sample points in a three-dimensional scene within a field of view of the multiple camera rig; setting the convergence distance to be an average sample point depth of the plurality of sample point depths; determining a minimum sample point depth of the plurality of sample point depths; and setting the baseline to be a negative inverse of a difference between the minimum sample point depth and the convergence distance.
2. The method of setting a baseline and a convergence distance of claim 1, wherein determining the plurality of sample point depths comprises performing ray casting within the three-dimensional scene and recording hit distances for each sample point of the plurality of sample points.
3. The method of setting a baseline and a convergence distance of claim 1, wherein determining the plurality of sample point depths comprises computing the sample point depths from a disparity map of the three-dimensional scene for each sample point of the plurality of sample points.
4. The method of setting a baseline and a convergence distance of claim 3, wherein computing the sample point depths further comprises using image disparity between images captured by different cameras of the multiple camera rig and forming a disparity map of the three-dimensional scene from the image disparity.
5. The method of setting a baseline and a convergence distance of claim 1, wherein determining the plurality of sample point depths comprises using a depth sensor to measure distances between the multiple camera rig and objects within the three-dimensional scene, the objects corresponding to sample points of the plurality of sample points.
6. The method of setting a baseline and a convergence distance of claim 5, wherein the depth sensor comprises one of a laser distance sensor and a time-of-flight distance sensor.
7. The method of setting a baseline and a convergence distance of claim 1, wherein the average sample point depth is a weighted average having weights assigned according to a decreasing

function of distance from a focal point within the three-dimensional scene.

8. The method of setting a baseline and a convergence distance of claim 7, wherein the weights of the weighted average are assigned according to a gaussian distribution centered on the focal point.

9. The method of setting a baseline and a convergence distance of claim 1, wherein the multiple camera rig comprises a plurality of virtual cameras, the three-dimensional scene being a three-dimensional model.

10. The method of setting a baseline and a convergence distance of claim 1, wherein the multiple camera rig comprises a plurality of physical cameras, the three-dimensional scene representing a physical scene imaged by cameras of the plurality of physical cameras.

11. The method of setting a baseline and a convergence distance of claim 1, wherein the multiple camera rig comprises a camera that is moved between a plurality of locations to capture images that form the three-dimensional scene.

12. A light field camera system comprising: a multiple camera rig comprising a plurality of cameras; a processor; and a memory configured to store instructions that when executed by the processor implement: determining a plurality of sample point depths representing distances between the multiple camera rig and a plurality of sample points in a three-dimensional scene within a field of view of the multiple camera rig; and setting a convergence distance and a baseline of the multiple camera rig, wherein the convergence distance is an average sample point depth of the plurality of sample point depths and the baseline is a negative inverse of a minimum sample point depth of the plurality of sample point depths minus the convergence distance.

13. The light field camera system of claim 12, wherein the plurality of sample point depths is determined one or more of from a depth map of an image representing the three-dimensional scene, using a disparity map to compute the sample point depths, performing ray casting within the three-dimensional scene to determine the sample point depths within the three-dimensional scene.

14. The light field camera system of claim 12, wherein the plurality of sample point depths is determined using a depth sensor to measure distances between the multiple camera rig and objects within the three-dimensional scene, the objects corresponding to sample points of the plurality of sample points.

15. The light field camera system of claim 12, wherein the average sample point depth of the plurality of sample point depths is a weighted average, weights of the weighted average being assigned according to a decreasing function of distance from a focal point of the three-dimensional scene.

16. The light field camera system of claim 12, wherein a camera of the plurality of cameras is a virtual camera, the three-dimensional scene being a three-dimensional model.

17. The light field camera system of claim 12, wherein cameras of the plurality of cameras of the multiple camera rig comprises physical cameras, the three-dimensional scene representing a physical scene imaged by the physical cameras.

18. The light field camera system of claim 12, further comprising a multiview display, the convergence distance corresponding to a zero disparity plane of the multiview display.

19. A non-transitory, computer-readable storage medium storing executable instructions that, when executed by a processor of a computer system, performs operations of determining a baseline and convergence distance of a multiple camera rig, the operations comprising: determining a plurality of sample point depths representing distances between the multiple camera rig and a plurality of sample points in a three-dimensional scene within a field of view of the multiple camera rig, the convergence distance being set as an average sample point depth of the plurality of sample point depths; and determining a minimum sample point depth of the plurality of sample point depths, the baseline being set as a negative inverse of a difference between the minimum sample point depth and the convergence distance.

20. The non-transitory, computer-readable storage medium of claim 19, wherein determining the plurality of sample point depths comprises one or more of from a depth map of an image

representing the three-dimensional scene, using a disparity map to compute the sample point depths, performing ray casting within the three-dimensional scene to determine the sample point depths within the three-dimensional scene, and using a depth sensor to measure distances between the multiple camera rig and objects within the three-dimensional scene, the objects corresponding to sample points of the plurality of sample points.

21. The non-transitory, computer-readable storage medium of claim 19, wherein the average sample point depth of the plurality of sample point depths is a weighted average, weights of the weighted average being assigned according to a decreasing function of distance from a focal point of the three-dimensional scene.
